

FIFA World Cup Winner Prediction using Machine Learning

FIFA World Cup Winner Prediction Report

This report presents a complete data science pipeline for predicting FIFA World Cup winners using machine learning techniques. The project involves data cleaning, feature engineering, model training, evaluation, and visualization via an interactive Streamlit app.

1. Introduction

Football, also known as soccer, is one of the most popular sports in the world. Predicting the outcome of major tournaments such as the FIFA World Cup has always been a challenging and exciting task.

This project applies machine learning techniques to historical World Cup match data to predict potential winners and top-performing teams for future tournaments.

2. Data Cleaning

The dataset consists of historical FIFA World Cup matches, containing team names, goals scored, and match outcomes. The cleaning process involved:

- Standardizing column names (HomeTeam, AwayTeam, HomeGoals, AwayGoals, Year).
- Removing duplicates and invalid entries.
- Extracting winners from each match based on goals scored.
- Handling missing values and ensuring numeric goal data types.

The cleaned dataset is stored as 'cleaned_fifa.csv'.

3. Feature Engineering

Team-level statistics are generated for each tournament year, including:

- Matches played, wins, draws, and losses.
- Goals for and against, goal difference.

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- Win rate and average goals per match.
- Past tournament appearances as an experience proxy.

These aggregated statistics are used as model input features.

4. Model Training and Evaluation

Two classification models were trained: Logistic Regression and Random Forest.

The target variable was whether a team won the tournament (`is_winner = 1`).

Evaluation metrics include Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Feature importance plots, confusion matrices, and ROC curves were generated to compare model performance.

5. Prediction and Visualization

The trained models predict the probability of each team winning the World Cup.

The app displays the top 10 predicted teams and the most likely winner for the selected year.

Visualizations include bar charts of predicted probabilities, scatter plots (goal difference vs win rate), and historical winners by year.

6. Streamlit Application Code

The following Python code implements the Streamlit web app for this project.

It includes modules for file upload, data preprocessing, model training, prediction, and visualization.

```
# fifa_worldcup_app.py
import streamlit as st
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestClassifier
# ...
# Full code implements:
# - Data cleaning and CSV export
# - Team-level feature engineering
# - Model training (LR & RF)
# - Metrics, ROC, and visualizations
```

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```
# - Top-10 team prediction display  
# - Download options for all outputs
```

7. Conclusion

This project demonstrates how machine learning and data visualization can be applied to sports analytics.

While predictions depend on historical performance, incorporating current player statistics and FIFA rankings

can further improve accuracy. The Streamlit app provides an interactive environment for exploring data, training models, and making predictions.